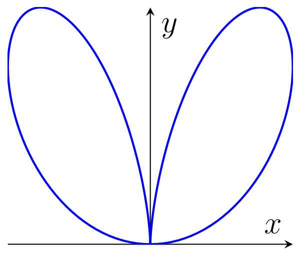
2025-2026 Spring
MAT124 Midterm
17/04/2026

1. A point is given in cylindrical coordinates as $(r, \theta, z) = \left(\sqrt{3}, -\pi/4, -1\right)$. What are its spherical coordinates (ρ, θ, ϕ)?

(A) $(2, -\pi/4, 2\pi/3)$ (B) $(2, \pi/4, \pi/3)$ (C) $(4, -\pi/4, 5\pi/6)$
(D) $(2, -\pi/4, \pi/6)$ (E) $(\sqrt{2}, -\pi/4, 3\pi/4)$

2. Which of the following polar equations represents a curve that has loops oriented primarily along the y-axis, as shown in the figure?



(A) $r = \cos^2\theta \sin\theta$ (B) $r = \sin^2\theta\cos\theta$ (C) $r = 1 + \sin\theta$
(D) $r = \sin(2\theta)$ (E) $r = 3\cos\theta$

3. Find the projection of the point $P(1, 2, 3)$ onto the plane $x + y + z = 0$.

(A) $(-1, 0, 1)$ (B) $(1, 1, -2)$ (C) $(0, 0, 0)$ (D) $(-1, -1, 2)$ (E) $(1, 2, -3)$

4. Find the area of the region inside the circle $r = 3\cos\theta$ and outside the cardioid $r = 1 + \cos\theta$.

(A) π (B) $\pi/2$ (C) 2π (D) $3\pi/4$ (E) 1

5. Identify the quadric surface defined by the equation $2z^2 - x^2/2 - y^2 - 5 = 0$.

(A) Hyperboloid of two sheets (B) Hyperboloid of one sheet
(C) Elliptic paraboloid (D) Cone (E) Cylinder

6. In which direction does $f(x, y, z) = x^2 + 2y^2 - 3z$ increase most rapidly at $P(1, 1, 2)$?

(A) $\langle 2, 4, -3\rangle$ (B) $\langle 1, 1, -3\rangle$ (C) $\langle -2, -4, 3\rangle$ (D) $\langle 2, 4, 0\rangle$ (E) $\langle 1, 2, -3\rangle$

7. Let $w = \ln(x^2 + y^2)$ where $x = s\cos t$ and $y = s + \sin t$. Find $\dfrac{\partial w}{\partial t}$ at $(s, t) = (1, \pi/6)$.

(A) 0 (B) 1 (C) $3/2$ (D) $\sqrt{3}/2$ (E) $\sqrt{3}/3$

8. Find the arc length of the curve $y = x^2/2, z = x^3/6$ from $x = 0$ to $x = 2$.

(A) $10/3$ (B) $8/3$ (C) 4 (D) $7/3$ (E) 3

9. A vector parallel to the tangent line of the intersection of $z = x^2+2y^2$ and $x^2+(y-1)^2 = 1$ at $P(1,1,3)$ is:

(A) $\langle 0,2,8\rangle$ (B) $\langle 2,4,-1\rangle$ (C) $\langle 2,0,0\rangle$ (D) $\langle 1,1,3\rangle$ (E) $\langle 4,-2,-8\rangle$

10. Which is a valid parametrization for the intersection of $z = x^2 + y^2$ and $z = 4$?

(A) $\vec{r}(t) = \langle 2\cos t, 2\sin t, 4\rangle$ (B) $\vec{r}(t) = \langle \cos t, \sin t, 4\rangle$

(C) $\vec{r}(t) = \langle 2\cos t, 2\sin t, t\rangle$ (D) $\vec{r}(t) = \langle t, t^2, 4\rangle$ (E) $\vec{r}(t) = \langle 2\cos t, 2\sin t, 2\rangle$

11. Consider the system of equations:

$$\begin{cases} xu^2v + y\cos(u) = 1 \\ \ln(uv) + x^2y - u + v = 0 \end{cases}$$

Assume that this system defines u and v implicitly as functions of x and y near the point $P(x,y,u,v) = (1,0,1,1)$. Show that the system can be solved for u and v in terms of x and y near P and find the value of $\partial u/\partial y$ at P.

12. **a.** Consider the function

$$f(x,y) = \begin{cases} \frac{x^2+\sin(x)\cos(y)}{x}, & x \neq 0 \\ 1, & x = 0 \end{cases}.$$

Find $f_x(0,0)$ using the formal definition of the partial derivative.

b. Evaluate the following limit or show that it does not exist.

$$\lim_{(x,y)\to(0,0)} \frac{x^3y^2}{x^6+y^4}$$

13. Find the absolute maximum and minimum values of the function

$$f(x,y) = x^2 + y^2 - 2x + 4y$$

on the closed triangular region D with vertices $(0,0)$, $(4,0)$, and $(0,-4)$. Show all your work, including the analysis of the interior critical points and the boundary of the region.

1. Given cylindrical coordinates $(r, \theta, z) = \left(\sqrt{3}, -\frac{\pi}{4}, -1\right)$. Using conversion formulas:

$$\begin{aligned}
\rho &= \sqrt{r^2 + z^2} = \sqrt{(\sqrt{3})^2 + (-1)^2} = \sqrt{3+1} = 2 \\
\theta &= -\frac{\pi}{4} \\
\phi &= \arccos\left(\frac{z}{\rho}\right) = \arccos\left(-\frac{1}{2}\right) = \frac{2\pi}{3}
\end{aligned}$$

Thus $(\rho, \theta, \phi) = \left(2, -\frac{\pi}{4}, \frac{2\pi}{3}\right)$.

$\boxed{\text{Answer: (A)}}$

2. We can solve this question by trial and error. Inspect $r = \cos^2\theta \sin\theta$.

- When $\theta = 0, \pi$, $r = 0$.
- For $\theta \in (0, \pi)$, $\sin\theta > 0$ and $\cos^2\theta \geq 0$, creating a loop along the positive y-axis.
- The factor $\cos^2\theta$ pinches the loop at $\theta = \frac{\pi}{2}$, matching the given shape.

$\boxed{\text{Answer: (A)}}$

3. Given point $P(1, 2, 3)$ and plane $x + y + z = 0$ with normal vector $\vec{n} = \langle 1, 1, 1\rangle$.

The orthogonal line through P is $\vec{r}(t) = \langle 1 + t, 2 + t, 3 + t\rangle$. Substitute into the plane equation.

$$(1 + t) + (2 + t) + (3 + t) = 0 \implies 3t + 6 = 0 \implies t = -2$$

Substitute $t = -2$ back yields the projection point.

$$P' = (1 - 2, 2 - 2, 3 - 2) = (-1, 0, 1)$$

$\boxed{\text{Answer: (A)}}$

4. Find intersection points for $r_1 = 3\cos\theta$ and $r_2 = 1 + \cos\theta$.

$$3\cos\theta = 1 + \cos\theta \implies 2\cos\theta = 1 \implies \theta = \pm\frac{\pi}{3}$$

Calculate area.

$$\begin{aligned}
A = \frac{1}{2}\int_{-\pi/3}^{\pi/3} \left(r_1^2 - r_2^2\right) d\theta &= \int_0^{\pi/3} \left(8\cos^2\theta - 2\cos\theta - 1\right) d\theta \\
&= \int_0^{\pi/3} (3 + 4\cos 2\theta - 2\cos\theta)\, d\theta \\
&= [3\theta + 2\sin 2\theta - 2\sin\theta]_0^{\pi/3} = \pi
\end{aligned}$$

$\boxed{\text{Answer: (A)}}$

5. Rearrange $2z^2 - \frac{x^2}{2} - y^2 - 5 = 0$.

$$\frac{z^2}{5/2} - \frac{x^2}{10} - \frac{y^2}{5} = 1$$

The standard form $\frac{z^2}{c^2} - \frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ has two negative minus signs and one positive sign, representing a hyperboloid of two sheets.

$\boxed{\text{Answer: (A)}}$

6. The direction of maximum increase of $f(x, y, z) = x^2 + 2y^2 - 3z$ is given by gradient ∇f:

$$\nabla f = \langle 2x, 4y, -3\rangle$$

Evaluate at $P(1, 1, 2)$.

$$\nabla f(1, 1, 2) = \langle 2, 4, -3\rangle$$

$\boxed{\text{Answer: (A)}}$

7. Given $w = \ln(x^2 + y^2)$, $x = s\cos t$, and $y = s + \sin t$. At $(s, t) = \left(1, \frac{\pi}{6}\right)$:

$$x = \frac{\sqrt{3}}{2}, \quad y = \frac{3}{2}, \quad x^2 + y^2 = 3$$

Applying the Chain Rule $\frac{\partial w}{\partial t} = \frac{\partial w}{\partial x}\frac{\partial x}{\partial t} + \frac{\partial w}{\partial y}\frac{\partial y}{\partial t}$:

$$\frac{\partial w}{\partial x} = \frac{2x}{x^2 + y^2} = \frac{\sqrt{3}}{3}, \qquad \frac{\partial x}{\partial t} = -s\sin t = -\frac{1}{2}$$

$$\frac{\partial w}{\partial y} = \frac{2y}{x^2 + y^2} = 1, \qquad \frac{\partial y}{\partial t} = \cos t = \frac{\sqrt{3}}{2}$$

$$\frac{\partial w}{\partial t} = \left(\frac{\sqrt{3}}{3}\right)\left(-\frac{1}{2}\right) + (1)\left(\frac{\sqrt{3}}{2}\right) = \frac{\sqrt{3}}{3}$$

$\boxed{\text{Answer: (E)}}$

8. For $y = \frac{x^2}{2}$ and $z = \frac{x^3}{6}$, derivatives are $\frac{dy}{dx} = x$ and $\frac{dz}{dx} = \frac{x^2}{2}$.

$$L = \int_0^2 \sqrt{1 + \left(\frac{dy}{dx}\right)^2 + \left(\frac{dz}{dx}\right)^2}\, dx = \int_0^2 \sqrt{1 + x^2 + \frac{x^4}{4}}\, dx$$

$$= \int_0^2 \left(1 + \frac{x^2}{2}\right) dx = \left[x + \frac{x^3}{6}\right]_0^2 = \frac{10}{3}$$

$\boxed{\text{Answer: (A)}}$

9. For $F(x,y,z) = x^2 + 2y^2 - z = 0$ and $G(x,y,z) = x^2 + (y-1)^2 - 1 = 0$, gradients at $P(1,1,3)$ are:

$$\nabla F(1,1,3) = \langle 2,4,-1\rangle, \quad \nabla G(1,1,3) = \langle 2,0,0\rangle$$

Cross product gives the tangent vector.

$$\vec{v} = \nabla F \times \nabla G = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 2 & 4 & -1 \\ 2 & 0 & 0 \end{vmatrix} = \langle 0,-2,-8\rangle$$

Negating yields the parallel vector $\langle 0,2,8\rangle$.

$\boxed{\text{Answer: (A)}}$

10. Intersecting $z = x^2 + y^2$ and $z = 4$ gives $x^2 + y^2 = 4$ in the plane $z = 4$.

Standard parametrization is

$$\vec{r}(t) = \langle 2\cos t, 2\sin t, 4\rangle$$

$\boxed{\text{Answer: (A)}}$

11. Jacobian matrix with respect to (u,v) at $P(1,0,1,1)$:

$$J = \begin{bmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{bmatrix}_{\text{at } P} = \begin{bmatrix} 2xuv - y\sin u & xu^2 \\ \frac{1}{u} - 1 & \frac{1}{v} + 1 \end{bmatrix}_{\text{at } P} = \begin{bmatrix} 2 & 1 \\ 0 & 2 \end{bmatrix}$$

Since $\det(J) = 2\cdot 2 - 1\cdot 0 = 4 \neq 0$, the system defines u, v implicitly near P. Differentiate with respect to y.

$$2u\frac{\partial u}{\partial y} + \frac{\partial v}{\partial y} + \cos(1) = 0 \implies 2\frac{\partial u}{\partial y} + \frac{\partial v}{\partial y} = -\cos(1)$$
$$2\frac{\partial v}{\partial y} + 1 = 0 \implies \frac{\partial v}{\partial y} = -\frac{1}{2}$$

Substitute $\frac{\partial v}{\partial y}$.

$$\boxed{\frac{\partial u}{\partial y} = \frac{1 - 2\cos(1)}{4}}$$

12. **a.** Use the formal definition of $f_x(0,0)$.

$$f_x(0,0) = \lim_{h\to 0} \frac{f(h,0) - f(0,0)}{h} = \lim_{h\to 0} \frac{\frac{h^2+\sin(h)}{h} - 1}{h} = \lim_{h\to 0} \frac{h^2 + \sin(h) - h}{h^2}$$

Use L'Hôpital's rule twice.

$$f_x(0,0) \overset{\text{L'H.}}{=} \lim_{h\to 0} \frac{2h+\cos h - 1}{2h} \overset{\text{L'H.}}{=} \lim_{h\to 0} \frac{2-\sin h}{2} = \frac{2-0}{2} = \boxed{1}$$

b. Approach $\displaystyle\lim_{(x,y)\to(0,0)} \frac{x^3y^2}{x^6+y^4}$ along paths $y = kx^{3/2}$.

$$\lim_{x\to 0^+} \frac{x^3(k^2x^3)}{x^6+k^4x^6} = \frac{k^2}{1+k^4}$$

Since the result depends on k, $\boxed{\text{the limit does not exist}}$.

13. The extreme value theorem states that a continuous function on a specific region will always have its absolute maxima and minima. f is continuous everywhere on its domain. Check the interior points.

$$\nabla f = \langle 2x-2, 2y+4\rangle = \langle 0,0\rangle \implies (1,-2)$$

Check if the point is inside the region.

$$y > x-4 \longrightarrow 1-(-2) = 3 < 4$$

$(1,-2)$ is inside. We have $f(1,-2) = -5$.

Check the boundary.

- $x = 0$: $g(y) = y^2+4y \implies y = -2 \implies f(0,-2) = -4$
- $y = 0$: $h(x) = x^2-2x \implies x = 1 \implies f(1,0) = -1$
- $y = x-4$: $k(x) = 2x^2-6x \implies x = \frac{3}{2},\ y = -\frac{5}{2} \implies f\left(\frac{3}{2}, -\frac{5}{2}\right) = -4.5$

Check the endpoints. At endpoints: $f(0,0) = 0$, $f(0,-4) = 0$, and $f(4,0) = 8$.

Compare all the values obtained.

$\boxed{\text{Absolute minimum value is } -5 \text{ at } (1,-2). \text{ Absolute maximum value is } 8 \text{ at } (4,0).}$